

# Navigating by Falling Stars

## Monetary Policy with Fiscally Driven Natural Rates

based on Campos, Fernández-Villaverde, Nuño & Paz (2026)

Warsaw Reading Group, July 2026

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# The classical models

In the canonical RANK model, the central bank hits its target if:

- 1 Fiscal policy stabilizes debt
- 2 Monetary policy follows the Taylor principle
- 3 The central bank tracks the natural rate  $r^*$ , which depends only on preferences/technology (e.g. the discount factor)

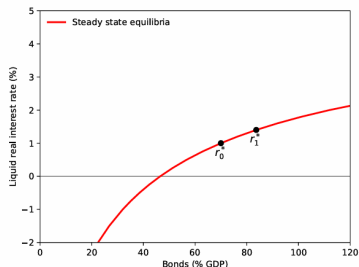
Crucially: in RANK,  $r^*$  is independent of fiscal policy.

# Why this breaks down in HANK

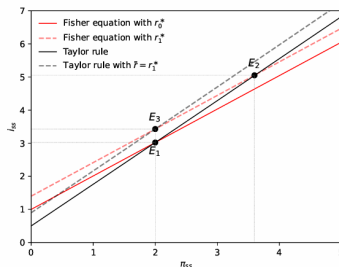
- In heterogeneous-agent New Keynesian (HANK) models, market incompleteness means households self-insure using government bonds.
- The stock of public debt determines how much insurance is available  
⇒ it moves the interest rate at which the bond market clears
- Fiscal policy therefore shifts the natural rate — unlike in RANK.
- This is not the FTPL: here the treasury still stabilizes the long-run real stock of debt (a passive fiscal rule). The channel is that fiscal policy shifts  $r^*$ , not that it abandons debt sustainability.

# How debt pins down $r^*$

Households' precautionary saving generates an upward-sloping bond demand curve; the treasury's budget constraint sets bond supply.



(a) Determination of  $r^*$



(b) Steady-state Taylor rule and Fisher equation

*Note:* Bonds (%GDP) denotes the real market value of government debt relative to annual GDP, expressed in percentage points.

# Long-run inflation and the policy trade-off

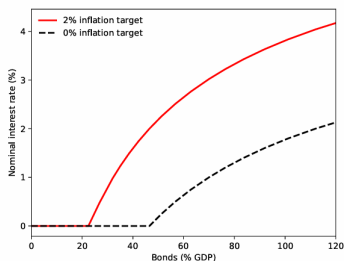
Approximating the Taylor rule in steady state and combining with the Fisher equation:

$$\pi_{ss} \approx \pi + \frac{r^* - r}{\phi_\pi - 1}$$

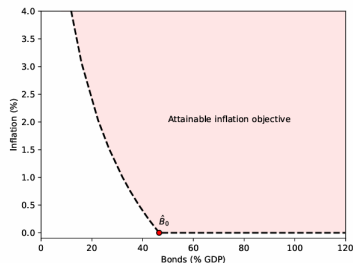
If the central bank does not update the rule intercept  $r$  when  $r^*$  rises (due to higher debt), steady-state inflation overshoots the target.

# The ZLB sets a minimum debt level

- If  $r^*(B_{ss}) + \pi \leq 0$ , the ZLB binds even after the central bank adjusts  $r$ :  $i_{ss} = 0$  and  $\pi_{ss} = -r^*(B_{ss})$ .
- There exists a minimum debt level  $\hat{B}_\pi$  below which the inflation target is unattainable.



(a) Steady-state nominal interest rate



(b) Steady-state inflation

# A debt-financed fiscal expansion

Transition from 70% to 80% debt-to-GDP, financed by government consumption (alternatively: taxes or transfers):

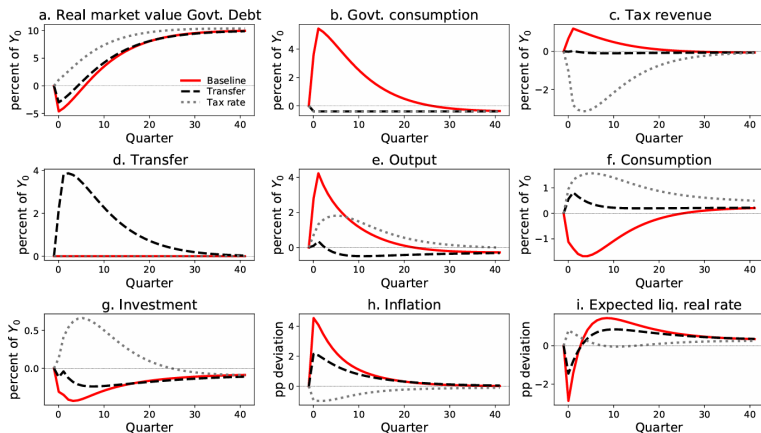
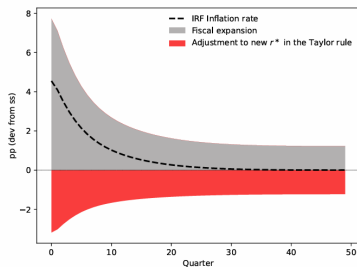


Figure 4: Comparison of different dynamics after a surprise debt-financed fiscal expansion

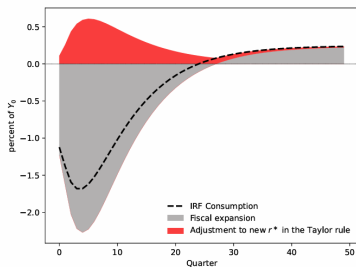


# Decomposing the transmission: fiscal vs. monetary

Decompose IRFs into a fiscal component (debt expansion, Taylor rule intercept fixed) and a monetary component (the intercept's adjustment to  $r^*$ ). Trade-off: track  $r^*$  closely  $\Rightarrow$  better price stability but a smaller fiscal multiplier; ignore it  $\Rightarrow$  larger multiplier but persistently higher inflation.



(a) Inflation



(b) Consumption

Figure 5: Decomposition of the IRFs of inflation and consumption

# Rules that don't need to know $r^*$

Alternative: the Orphanides–Williams (2002) differential rule,

$$\log(1 + i_t) = \log(1 + i_{t-1}) + \phi_{\pi}^{OW} \log\left(\frac{1+\pi_t}{1+\pi}\right),$$

Delivers markedly smoother inflation/rate paths and earlier consumption recovery than the standard Taylor rule; history dependence, not extra target variables, seems to be what matters.

Comparison with Holden's (2024) robust real rate rule: sharper short-run declines in consumption/investment, since inflation is kept exactly at target and debt is not diluted by surprise inflation.

# Does the timing of the policy response matter?

Compare central bank adjusting  $r$  to the new  $r^*$ : (i) immediately upon announcement of a future fiscal expansion, (ii) at implementation, or (iii) with a 12-quarter delay.

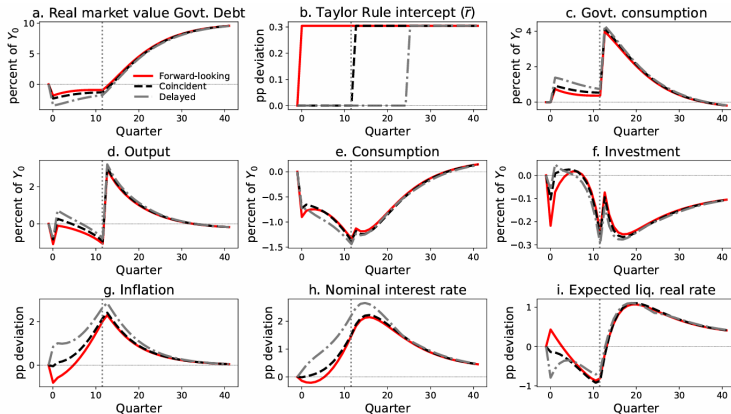


Figure 7: Dynamics of an anticipated debt-financed fiscal expansion

# Conclusion

- In a quantitative two-asset HANK model, public debt shifts the natural interest rate — a channel absent from RANK models.
- Price stability requires the central bank to adjust its policy-rule intercept in step with the fiscal stance; failing to do so raises steady-state inflation, government also takes into account ZLB.
- Fiscal expansions generate sizable short-run inflation, with dynamics that depend heavily on which instrument finances them and when monetary policy responds.
- Rules that sidestep the need to estimate  $r^*$  (e.g. Orphanides–Williams or Holden) can deliver superior stabilization.